

# SUHAS S

Bengaluru, India | +91 8867406327 | rajsuhas780@gmail.com

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## EXPERIENCE

### Freo

Jan 2024 – Present

*Software Engineer (Backend)*

*Bengaluru, India*

- **Led** migration of the core **Loan Management System** from proprietary Finflux to open-source **Apache Fineract** via a decoupled, **shadow-mode, transformer-based architecture**; ensured backward compatibility, zero downtime and future LMS portability while **eliminating ~Rs.40L/month in licensing cost**.
- **Designed** and delivered end-to-end integrations for **2 lending partners** across onboarding, loan creation, repayments and foreclosure – handling each partner’s distinct API contracts and business rules – **enabling Rs.3Cr+ in monthly disbursements** with fault-tolerant, highly available transaction handling.
- **Architected** a scalable, **event-driven VKYC workflow** with **Kafka-based async processing**, third-party vendor integration and high-throughput REST APIs; applied the **Strategy pattern** for partner-specific KYC routing, **retaining a key partner disbursing Rs.1Cr+/month** that set VKYC as a condition to continue.
- **Optimized** the high-traffic **customer data API** by eliminating an **N+1 query pattern** (replacing per-loan fetches with a **single aggregate query**), adding **Redis caching**, parallelizing calls and moving partner-API calls off the request path – **cutting latency from ~900 ms to under 100 ms**.
- **Engineered** the **Repayments V2** module with **idempotent retry** and automated **settlement reconciliation**, **reducing repayment-related support tickets by ~75%**.
- **Integrated** the **PayU payment gateway** as a lower-cost alternative to Razorpay and re-architected the repayment collection flow, **reducing per-transaction processing fees (~33% lower on UPI)** and cutting monthly payment-processing spend.
- **Launched** a **repayment-time cross-sell module** bundling optional add-ons (insurance, credit-report renewals) onto customer EMIs at checkout, using extensible **design patterns**, **contributing ~Rs.25L/month in incremental revenue**.

## TECHNICAL SKILLS

**Languages:** Java, SQL, Python

**Frameworks:** Spring Boot, Spring Data JPA, Hibernate, Spring AOP, Microservices, REST APIs

**Databases:** MySQL, PostgreSQL, Redis, DynamoDB, Aurora RDS

**Messaging & Streaming:** Apache Kafka, AWS SQS

**Cloud & DevOps:** AWS (ECS, S3, Lambda, API Gateway, CloudWatch, CodeArtifact, ELB), CI/CD, Git, Maven

**Monitoring:** SigNoz, Sentry, Sumo Logic

**Concepts:** System Design, Distributed Systems, Caching, Event-Driven Architecture, Design Patterns, OOP, SOLID, Concurrency, Scalability, Data Structures & Algorithms

## PROJECTS

**Field Encryption SDK** | *Java 17, Spring Boot, Spring AOP, JPA, JUnit, Maven* | [GitHub](#)

2026

- **Built** a reusable Spring Boot SDK exposing a custom **@EncryptDecrypt** annotation that, via **Spring AOP**, transparently encrypts sensitive fields before persistence and decrypts them on retrieval – keeping crypto out of business logic.
- **Implemented AES-256-GCM** authenticated encryption with a random per-record IV and SHA-256 key derivation, covering round-trip, IV-uniqueness and tamper detection with **JUnit** tests.

**Distributed API Rate Limiter** | *Java 17, Spring Boot, Spring AOP, Redis, Lua* | [GitHub](#)

2026

- **Built** a distributed, **Redis-backed rate limiter** enforcing limits across stateless app servers via **atomic Lua scripts** that eliminate read-compute-write race conditions under concurrency.
- **Implemented** pluggable **Token Bucket**, **Sliding Window** and **Fixed Window** algorithms behind a **@RateLimit** annotation (Spring AOP), with **fail-open** resilience and standard 429 / Retry-After headers.

## EDUCATION

**M S Ramaiah University**

Dec 2020 – Jul 2024

*B.Tech, Computer Science Engineering* CGPA: 9.35/10

*Bengaluru, India*

## CERTIFICATIONS

Programming in Java, Operating System Fundamentals, and Database Management System – NPTEL, Indian Institute of Technology (IIT) Kharagpur